EXPERIMENT 1

Traditional Development vs Low-Code Development – Case Study

AIM

To study and compare **Traditional Software Development** and **Low-Code Development** based on development time, coding requirements, flexibility, cost, and ease of use through a case study.

OBJECTIVE

To understand the differences between traditional development and low-code development and identify the suitable approach for developing software applications.

INTRODUCTION

Software applications can be developed using different approaches. **Traditional development** requires programmers to write code manually using programming languages and frameworks. **Low-code development** uses visual interfaces, drag-and-drop components, and pre-built modules to create applications with minimal coding.

CASE STUDY

Consider the development of a **Student Management System** using both traditional development and low-code development approaches.

Traditional Development

In traditional development, the developer manually writes code for the frontend, backend, database connectivity, authentication, and other application features. It provides high flexibility and complete control over the application.

Low-Code Development

In low-code development, the application is created using visual development tools, pre-built components, drag-and-drop interfaces, and automated workflows. It reduces the amount of manual coding and allows applications to be developed quickly.

COMPARISON

Feature	Traditional Development	Low-Code Development
Coding Requirement	High	Minimal
Development Time	More	Less
Flexibility	Very High	Moderate
Technical Skill	Requires programming knowledge	Requires less coding knowledge
Customization	Highly customizable	Limited to available features
Cost	Generally higher	Generally lower
Maintenance	Requires developers	Easier maintenance

OBSERVATION

From the case study, it was observed that **traditional development** provides greater flexibility, customization, and control over the application. However, it requires more coding knowledge and development time.

Low-code development reduces manual coding and allows applications to be developed quickly using visual tools and pre-built components. It is suitable for rapid application development and simple business applications.

RESULT

Thus, the differences between **Traditional Development and Low-Code Development** were studied and compared successfully through a case study. It was observed that traditional development is suitable for highly customized applications, whereas low-code development is suitable for faster application development with minimal coding.